

Functions

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 3 # _____ A relation is _____.	ANSWER: 1 # _____ Given $f(x) = \begin{cases} 4x - 5, & x \leq 2 \\ x^2 - 5, & x > 2 \end{cases}$, find $f(3)$.
ANSWER: range # _____ Determine if the set $\{(2, 0), (-1, 1), (3, 4), (0, 7), (-2, 4), (-3, 0)\}$ represents a function.	ANSWER: $3x + 8$ # _____ Identify the domain of $h(x) = \sqrt{5 - x}$.
ANSWER: $[-3, 3) \cup (3, \infty)$ # _____ How many of the equations below represent a function? $x = 4$, $y = 3 x$, $y^2 = x + 2$, $x = 3 y$, $x^2 + xy + y^2 = 1$.	ANSWER: any set of ordered pairs # _____ A function is a special kind of relation where _____ have the same x-coordinate.
ANSWER: $(-5, 5)$ # _____ Given $f(x) = \frac{x - 3}{\sqrt{x + 3}}$, identify the domain.	ANSWER: all real numbers # _____ Given $f(x) = \sqrt{25 - x^2}$, identify the domain of f.

ANSWER: $(-\infty, 5]$

_____ Identify the domain of $g(x) = \sqrt[3]{5-x}$.

ANSWER: 4

_____ Given $f(x) = \begin{cases} 4x - 5, & x \leq 2 \\ x^2 + 1, & x > 2 \end{cases}$, find $f(2)$.

ANSWER: $(-\infty, 1) \cup (1, \infty)$ # _____ Given $f(x) = \frac{12}{ x }$, find $f(-6)$.	ANSWER: domain # _____ The y-values make up the _____.
ANSWER: $[0, 5]$ # _____ Given $f(x) = \frac{1}{\sqrt{25 - x^2}}$, identify the domain of f.	ANSWER: yes # _____ Determine the domain of the set $\{(2, 0), (-1, 1), (3, 4), (0, 7), (-2, 4), (-3, 0)\}$.
ANSWER: no # _____ Determine the domain of $g(x) = \frac{7x}{x - 1}$.	ANSWER: 2 # _____ Given $h(x) = 3x - 1$, find $h(x + 3)$.
ANSWER: $\{-3, -2, -1, 0, 2, 3\}$ # _____ Does the mapping diagram shown represent a function?	ANSWER: $(-3, \infty)$ # _____ Given $f(x) = \frac{\sqrt{x+3}}{x-3}$, identify the domain of f.
ANSWER: no two ordered pairs # _____ The x-values make up the _____.	ANSWER: $[-5, 5]$ # _____ Given $f(x) = \sqrt{25 - x^2}$, identify the range of f.